Algebra 1
Unit 13
Lesson 1 Practice

Name_Answer Key_________
Date____________________
Period__________________

The Yang family is adding a second floor to their home. They were approved by their bank for a loan of $\$100{,}000$ with a simple interest rate of 4.85%.

1. Write an equation that models the Yang's loan.

$A = 100{,}000(1 + .0485t)$

2. Rewrite the equation in slope-intercept form.

$A = 4850t + 100{,}000$

3. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	$100{,}000$	Initial loan amount
slope	$4{,}850$	Yearly interest

4. What is the total the Yang's will have to pay if they pay off the loan in 10 years?

$A = 4{,}850(10) + 100{,}000$

$48{,}500 + 100{,}000$

$\boxed{\$148{,}500}$

5. How many years did it take the Yang's to pay off the loan if the total they paid was $\$135{,}000$?

$$\begin{aligned} 135{,}000 &= 4{,}850t + 100{,}000 \\ -100{,}000 &\qquad\quad - 100{,}000 \end{aligned}$$

$$\frac{35{,}000}{4850} = \frac{4850t}{4850}$$

$7.2 \approx t$ $\boxed{\text{About } 7.2 \text{ years}}$

Mekhi just graduated college and is starting his first job. He is investing money into a savings account to plan for retirement. He has $\$15{,}000$ to invest in a savings account with a 3.25% APR.

6. Write an equation that models Mekhi's savings account.

$y = 15{,}000(1 + .0325t)$

7. Write the equation in slope-intercept form.

$y = 487.5t + 15{,}000$

8. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	$15{,}000$	Initial investment
slope	487.50	Interest per year

9. How much money will be in the savings account if Mekhi wants to retire 30 years later?

$15{,}000 + 487.5(30) = y$

$15{,}000 + 14{,}625 = y$

$\$29{,}625 = y$

10. How many years would Mekhi have to work to retire with $\$35{,}000$?

$35{,}000 = 15{,}000 + 487.5t$

$20{,}000 = 487.5t$

$41 \approx t$
years

Algebra 1
Unit 13
Lesson 2 Practice

Name_Answer Key________
Date___________________
Period_________________

Juan deposited $\$20{,}000$ in a savings account with a 4.75% APR compounded quarterly.

1. Write an equation that models the amount of money Juan has in the account.

$$A = 20{,}000\left(1 + \frac{.0475}{4}\right)^{4t}$$

2. What is the balance in the account after 8 years?

$$A = 20{,}000\left(1 + \frac{.0475}{4}\right)^{4 \cdot 8}$$

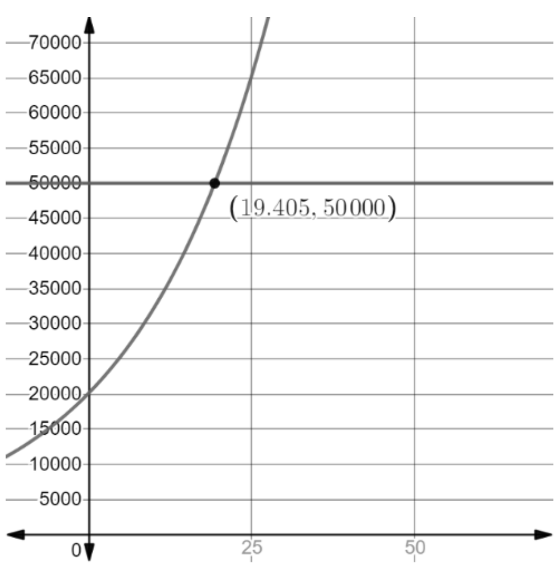
$$= \$29{,}180.30$$

3. What is the balance in the account after 12 years?

$$A = 20{,}000\left(1 + \frac{.0475}{4}\right)^{4 \cdot 12}$$

$$A = \$35{,}246.79$$

4. Use the graph to determine when the savings account will reach $\$50{,}000$.



After 19.4 years

For questions 5-10, complete the table below.

	Simple Interest	Compound Interest
Investment amount and APR	$150,000 is invested at an interest rate of 5.25% APR	$150,000 is invested at an interest rate of 5.25% APR compounded monthly
Write the equation.	5. $A = P(1 + rt)$ $A = 150{,}000(1 + .0525t)$	6. $A = P\left(1 + \frac{r}{n}\right)^{nt}$ $A = 150{,}000\left(1 + \frac{.0525}{12}\right)^{12t}$
Determine the amount after 5 years.	7. $A = 150{,}000(1 + .0525 \cdot 5)$ $A = \$189{,}375$	8. $A = 150{,}000\left(1 + \frac{.0525}{12}\right)^{12 \cdot 5}$ $A = \$194{,}914.84$
Determine the amount after 25 years.	9. $A = 150{,}000(1 + .0525 \cdot 25)$ $= \$346{,}875$	10. $A = 150{,}000\left(1 + \frac{.0525}{12}\right)^{12 \cdot 25}$ $A = \$555{,}724.43$

Algebra 1　　　　　　　　Name_Answer Key________
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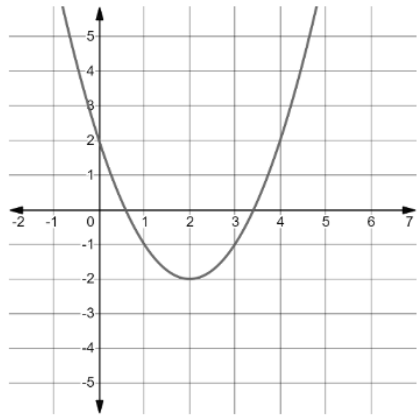
Lesson 3 Practice　　　　Period__________________

The table shows three different functions.

<table>
<tr><th>Function A</th><th>Function B</th><th>Function C</th></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>3</td></tr><tr><td>-1</td><td>6</td></tr><tr><td>0</td><td>9</td></tr><tr><td>1</td><td>12</td></tr><tr><td>2</td><td>15</td></tr></table></td><td></td><td>$y = 5(1.4)^x$</td></tr>
</table>

1. Determine which type of function that each represents.

Function A	Function B	Function C
o Exponential o Linear o Quadratic	o Exponential o Linear o Quadratic	o Exponential o Linear o Quadratic

2. Which function has the largest y-intercept?

Function A

3. Compare the ranges of the functions and indicate any similarities and differences.
Answers may vary.
$A \rightarrow -\infty < y < \infty$
$B \rightarrow y \geq -2$
$C \rightarrow y > 0$

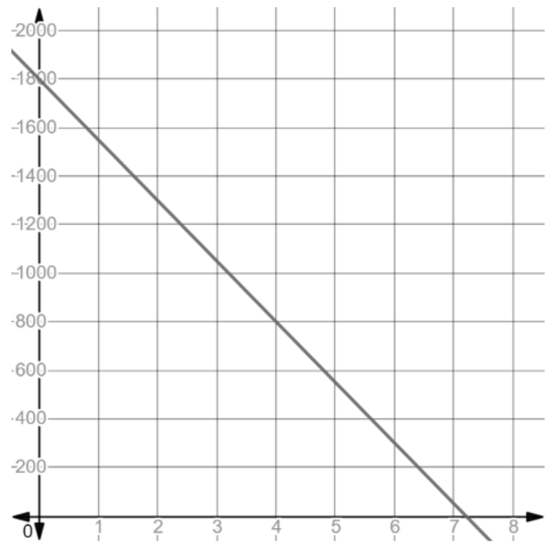
4. Compare the domains of the functions and indicate any similarities and differences.

All have a domain of all real numbers.

5. As $x \rightarrow \infty$, which function has larger values of y?

Function C

The table shows a table of values representing the revenue, in billions of dollars, from sales of digital movies, where $x = 0$ is 2010, and a graph representing the revenue, in billions of dollars, from sales of digital books, where $x = 0$ is 2010.

<table>
<tr><th>Digital Movie Sales
(billions of $)</th><th>Digital Book Sales
(billions of $)</th></tr>
<tr><td>
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>0</td><td>2057.35</td></tr>
<tr><td>1</td><td>1833.96</td></tr>
<tr><td>2</td><td>1634.83</td></tr>
<tr><td>3</td><td>1457.32</td></tr>
<tr><td>4</td><td>1299.08</td></tr>
</table>
</td><td>


</td></tr>
</table>

6. Approximate the x-intercept for the function that represents the sale of digital books.

   ≈ 7.3

7. Interpret the x-intercept of the function that represents the sale of digital books.

   Digital book sales would reach 0 in 2017.

8. What is the domain of the function representing the sale of digital books?

   0 to approximately 7.3
   $0 \leq x \leq 7.3$

9. Which function has a larger y-intercept?

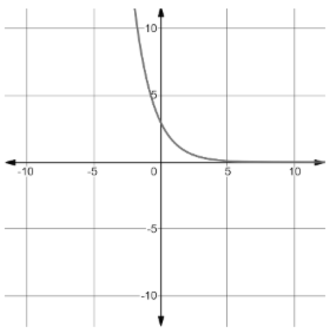
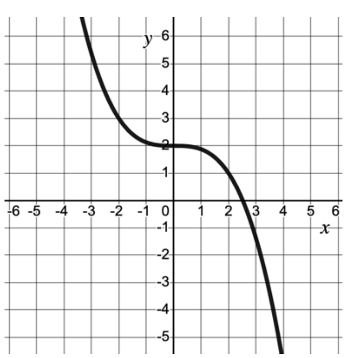
   Movie sales

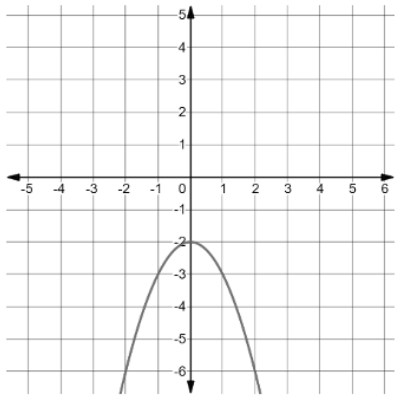
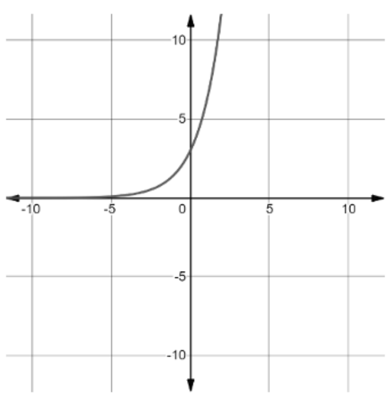
10. What type of function best describes the table of values that represents digital movie sales? Explain your choice. Explanations may vary.

    - Exponential
    - Linear
    - Quadratic

Algebra 1 Name_Answer Key________
Unit 13 Date____________________
Lesson 4 Practice Period__________________

Determine which family of functions each transformation is part of.

<table>
<tr><th>Representation of Transformation</th><th colspan="2">Equation of Parent Function</th></tr>
<tr><td>1.
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-2</td><td>-9</td></tr>
<tr><td>-1</td><td>-11</td></tr>
<tr><td>0</td><td>-13</td></tr>
<tr><td>1</td><td>-15</td></tr>
<tr><td>2</td><td>-17</td></tr>
</table></td>
<td>o $\boxed{y = x}$
o $y = x^2$
o $y = x^3$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $y = \frac{1}{2}^x$</td></tr>
<tr><td>2.

</td>
<td>o $y = x$
o $y = x^2$
o $y = x^3$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $\boxed{y = \frac{1}{2}^x}$</td></tr>
<tr><td>3.
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-2</td><td>-12</td></tr>
<tr><td>-1</td><td>-3</td></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>1</td><td>-3</td></tr>
<tr><td>2</td><td>-12</td></tr>
</table></td>
<td>o $y = x$
o $\boxed{y = x^2}$
o $y = x^3$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $y = \frac{1}{2}^x$</td></tr>
<tr><td>4.
$$y = \frac{2}{3}x + 6$$</td>
<td>o $\boxed{y = x}$
o $y = x^2$
o $y = x^3$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $y = \frac{1}{2}^x$</td></tr>
<tr><td>5.

</td>
<td>o $y = x$
o $y = x^2$
o $\boxed{y = x^3}$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $y = \frac{1}{2}^x$</td></tr>
</table>

<table>
<tr>
<td>6.
</td>
<td>o $y = x$
o $y = x^2$
o $\boxed{y = x^3}$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $y = \frac{1}{2}^x$</td>
</tr>
<tr>
<td>7.
<table><tr><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>y</td><td>18</td><td>6</td><td>2</td><td>$\frac{2}{3}$</td><td>$\frac{2}{9}$</td></tr></table></td>
<td>o $y = x$
o $y = x^2$
o $y = x^3$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $\boxed{y = \frac{1}{2}^x}$</td>
</tr>
<tr>
<td>8.
$y = \sqrt{x+6} - 4$</td>
<td>o $y = x$
o $y = x^2$
o $y = x^3$
o $\boxed{y = \sqrt{x}}$</td>
<td>o $y = |x|$
o $y = 2^x$
o $y = \frac{1}{2}^x$</td>
</tr>
<tr>
<td>9.
</td>
<td>o $y = x$
o $y = x^2$
o $y = x^3$
o $y = \sqrt{x}$</td>
<td>o $y = |x|$
o $\boxed{y = 2^x}$
o $y = \frac{1}{2}^x$</td>
</tr>
<tr>
<td>10.
$y = 2|x-5|$</td>
<td>o $y = x$
o $y = x^2$
o $y = x^3$
o $y = \sqrt{x}$</td>
<td>o $\boxed{y = |x|}$
o $y = 2^x$
o $y = \frac{1}{2}^x$</td>
</tr>
</table>

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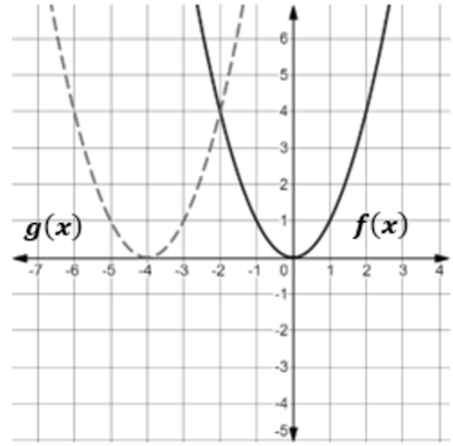
The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

   Shifts 4 units to the left

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1	1	-3	1
0	0	-4	0
1	1	-5	1

3. Explain the change in the x-values from $f(x)$ to $g(x)$.

   Subtract 4 to each x from $f(x)$ to $g(x)$.

4. Complete the statement:

   The graph of $g(x)$ is a
   - ✓ horizontal
   - o vertical

   shift of $f(x)$.

5. Complete the statement:

   The shift is seen in the table by
   - o adding
   - ✓ subtracting

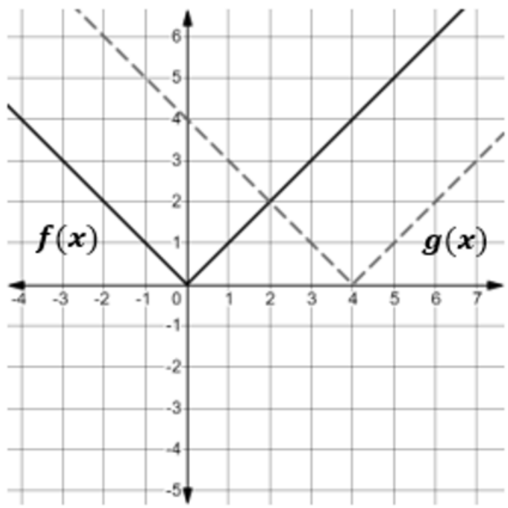
   4 to each
   - ✓ x-value
   - o y-value

   for each corresponding
   - o x-value.
   - ✓ y-value.

Complete the table for questions 6-8. The first row has been completed.

Original Equation	Original Function	Shift	Transformed Function
$y = x^2$	$f(x) = x^2$	Right 2	$f(x-2) = (x-2)^2$
6. $y = \lvert x \rvert$	$f(x) = \lvert x \rvert$	Right 4	$f(x-4) = \lvert x-4 \rvert$
7. $y = x^3$	$f(x) = x^3$	Left 4	$f(x+4) = (x+4)^3$
8. $y = \sqrt{x}$	$f(x) = \sqrt{x}$	Right 5	$f(x-5) = \sqrt{x-5}$

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



9. How does the graph of $f(x)$ compare to the graph of $g(x)$?

$g(x)$ is 4 units to the right of $f(x)$.

10. Two function descriptions are shown. Which function description matches the transformation on the graph? Justify your choice.

- o $g(x) = f(x+4)$
- o ✓ $g(x) = f(x-4)$

The vertex of $f(x)$ moved four units to the right resulting in $g(x)$.

Algebra 1
Unit 13
Lesson 6 Practice

Name_Answer Key________
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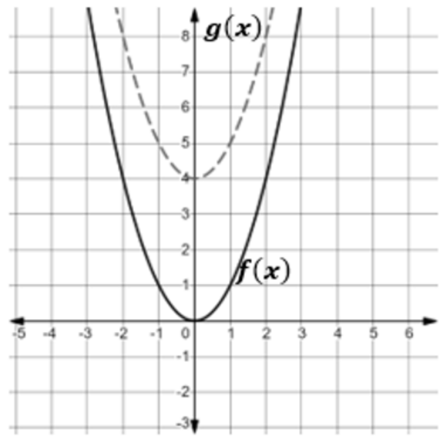
The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

   Shift up 4 units

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1	1	-1	5
0	0	0	4
1	1	1	5

3. Explain the change in the y-values from $f(x)$ to $g(x)$.

   y-values increase by 4.

4. Complete the statement:

   The graph of $g(x)$ is a
   - o horizontal
   - o vertical

   shift of $f(x)$.

5. Complete the statement:

   The shift is seen in the table by
   - o adding
   - o subtracting

   4 to each
   - o x-value
   - o y-value

   for each corresponding
   - o x-value.
   - o y-value.

For 6-7, complete the statements to describe how the graph of $g(x) = f(x) - 5$ compares to the graph of $f(x) = |x + 2|$.

6. The graph of $g(x)$ is a

- o horizontal
- o vertical

shift of $f(x)$.

7. The graph of $g(x)$ is shifted __5__ units

- o up from
- o down from
- o to the right of
- o to the left of

the graph of $f(x)$.

8. Describe the effect on the x-values of $K(x) = |x - 3|$ for the transformation $J(x) = K(x) + 6$.

y-values will be shifted up for 6 for each corresponding x-value.

9. Describe the effect on the y-values of $v(x) = \frac{3}{4}x + 7$ for the transformation $w(x) = v(x) - 3$.

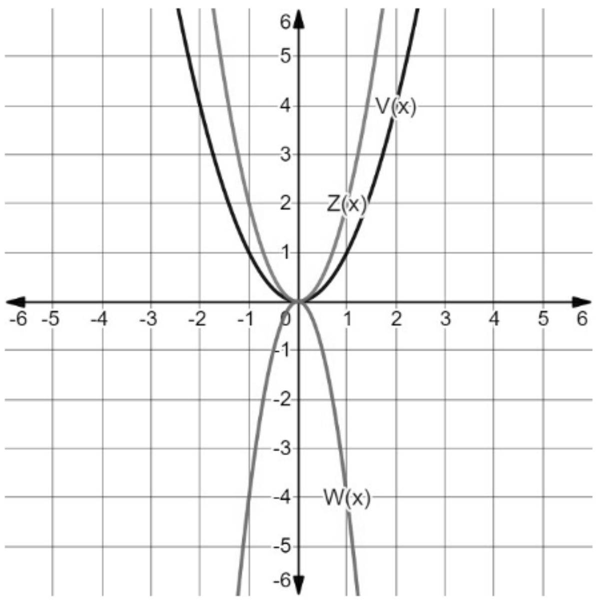
y-values will be shifted down 3 units.

10. Describe how the graph of $T(x) = S(x - 10)$ compares to the graph of $S(x) = x^2 + 2$.

$T(x)$ is shifted 10 units to the right.

Algebra 1 Name_Answer Key________
Unit 13 Date____________________
Lesson 7 Practice Period__________________

The functions $V(x)$, $W(x)$, and $Z(x)$ are shown on the graph, where $V(x) = x^2$, $W(x) = -4x^2$, and $Z(x) = 2x^2$. The table of values for each function are also shown.



$y = V(x)$		$y = W(x)$		$y = Z(x)$	
x	y	x	y	x	y
-3	9	-3	-36	-3	18
-2	4	-2	-16	-2	8
-1	1	-1	-4	-1	2
0	0	0	0	0	0
1	1	1	-4	1	2
2	4	2	-16	2	8
3	9	3	-36	3	18

Complete the statements below.

1. The coefficient of $V(x) = x^2$ is 1.
2. The coefficient of $W(x) = -4x^2$ is -4.
3. The y-values of $W(x)$ can be found by multiplying the y-values of $V(x)$ by -4.
4. The coefficient of $Z(x) = 2x^2$ is 2.
5. The y-values of $Z(x)$ can be found by multiplying the y-values of $V(x)$ by 2.

The table of values of the function $y = B(x)$ is shown.

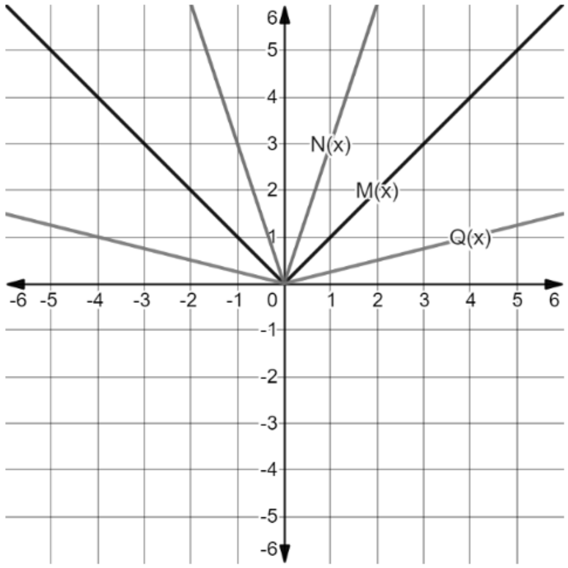
x	-6	-4	-2	0	2	4	6
y	4	7	10	13	16	19	22

6. Complete the sentence to describe the effect on the x-values for the function $y = B(2x)$. The x –values for $B(x)$ will be

o multiplied
✓ divided

by 2.

The functions $M(x)$, $N(x)$, and $Q(x)$ are shown on the graph, where $M(x) = |x|$, $N(x) = |3x|$, and $Q(x) = |0.25x|$. The table of values for each function are also shown.



$y = M(x)$		$y = N(x)$		$y = Q(x)$	
x	y	x	y	x	y
-2	2	$-\frac{2}{3}$	2	-8	2
-1	1	$-\frac{1}{3}$	1	-4	1
0	0	0	0	0	0
1	1	$\frac{1}{3}$	1	1	4
2	2	$\frac{2}{3}$	2	2	8

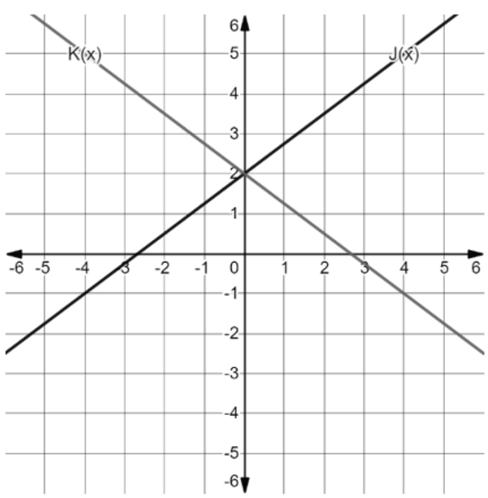
Complete the statements below.

7. The x-values of $N(x)$ can be found by [○ multiplying / ✓ dividing] the x-values of $M(x)$ by 3.

8. The x-values of $Q(x)$ can be found by [✓ multiplying / ○ dividing] the x-values of $M(x)$ by 4.

The functions $J(x)$ and $K(x)$ are shown on the graph where $K(x) = J(-x)$. The table of values for each function are also shown.

$y = J(x)$		$y = K(x)$	
x	y	x	y
-2	0.5	-2	3.5
-1	1.25	-1	2.75
0	2	0	2
1	2.75	1	1.25
2	3.5	2	0.5



9. What is the effect on the table of values when the x in $J(x)$ was replaced with $-x$ to create $K(x)$?

   As the x-values increase, the y-values decrease in $k(x)$.

10. What is the effect on the graph when the x in $J(x)$ was replaced with $-x$ to create $K(x)$?

    Reflection over the y-axis.

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Unit 13 Date___________________
Lesson 8 Practice Period__________________

Calista is running a marathon. Based on the temperature on race day and her average marathon time, she calculated that she needs to consume 120 fluid ounces throughout the race. She wants to use both sports drinks and plain water, which she will grab from aid stations along the course. She can get sports drink in 6-ounce cups and water in 8-ounce cups.

1. $6s + 8w$ is a(n)
   - o function.
   - o equation.
   - ★ expression.

2. The values 6 and 8 are
   - ★ coefficients.
   - o terms.
   - o constants.

3. The monomials $6s$ and $8w$ are
   - o coefficients.
   - ★ terms.
   - o constants.

4. $6s + 8w = 120$ is a(n)
   - o function.
   - ★ equation.
   - o expression.

5. Determine the amount of water Calista needs if she consumes 8 cups of sports drink.

$$6(8) + 8w = 120$$

$$48 + 8w = 120$$
$$-48 \qquad -48$$

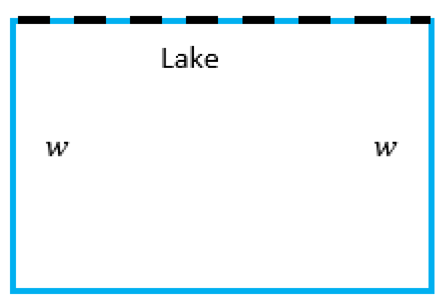
$$\frac{8w}{8} = \frac{72}{8}$$

$$w = 9$$

9 cups of water

A farmer has $1{,}000$ feet of fencing to enclose his property, which borders a lake. The diagram shows the width, w, of the property and the lake that his property borders. The fencing is not used along the lake.

Explain what each quantity represents in this context.



6. $1000 - 2w$
   The amount of fencing left after putting the fence on the left and right sides of the property. It is the length of the the property parallel to the lake.

7. $w(1000 - 2w)$
   The area of the farmer's property.

Midori's parents started a college savings fund when she was born in 2010. The amount of money in the savings fund since 2010 can be represented by the equation $y = 10{,}000(1 + 0.085)^x$, where y is the amount in the college fund and x is the number of years since 2010.

8. The equation $y = 10{,}000(1 + 0.085)^x$ represents

★ exponential growth.
o exponential decay.

9. The rate of

o decay
★ growth

is 8.5 %.

10. What does the coefficient $10{,}000$ represent?

The original amount of money deposited in 2010.

Algebra 1
Unit 13
Lesson 9 Practice

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To make a 14% acid solution, a chemist mixes 4 parts of 2% acid solution with an 18% acid solution. The expression $4(0.02x)+0.18y$ represents the mixture, where x is the number of ounces of 2% solution and y is the number of ounces of the 18% solution.

Explain what each quantity means.

1. $(0.02x)$

   Ounces of acid solution (2%).

2. $4(0.02x)$

   Amount of acid solution in mixture (2%).

3. $0.18y$

   Amount of acid solution (18%) in ounces.

Since 2018, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1{,}675{,}000(1.064)^x$.

4. What does the x variable represent?

   Years since 2018

5. What does the y variable represent?

   Number of independently contracted food delivery drivers since 2018.

6. What does the quantity $1{,}675{,}000$ represent?

   Initial amount of independently contracted drivers.

7. What is the constant percent rate of change?

   106.4% increase

The equation $y = -4(x-2)^2 + 20$ models the height, y, in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 2$

   The number of seconds away the ball is before reaching maximum height.

9. 20

   Maximum height of the ball, in feet.

Michael is buying coffee for $\$5.75$ per bag. He has a coupon for three free bags of coffee. He pays $\$25.50$ for a birthday cake. The expression $5.75(x-3) + 25.50$ can be used to represent the overall cost.

10. What does the quantity $5.75(x-3)$ represent?

    The amount paid for the total amount of coffee, in bags, obtained.

Algebra 1 Name_Answer Key_________
Unit 13 Date_____________________
Lesson 10 Practice Period___________________

Given a written description, identify the function as linear, quadratic, or exponential. Explain your choice.

1. An investment grows at a constant percent rate of 3.5% APR.

   Exponential: constant percent will always be exponential.

2. The Americans with Disabilities Act (ADA) requires wheelchair ramps to have a 1-inch rise for every 12 horizontal inches of ramp.

   Linear: constant ratio of $1:12$.

3. A ball is thrown from a height of 4 feet. It reaches a maximum height of 35 feet 3 seconds later.

   Quadratic: follows a quadratic curve as the distance from ground level increases, reaches its maximum, then changes directions back towards the ground.

4. A rare species of bird experiences a population decline at a constant percent rate of 3%.

   Exponential: constant percent will always be exponential.

Given a table of values, identify the function as linear, quadratic, or exponential. Explain your choice.

5.

x	y
1	34
2	46
3	50
4	46
5	34

Quadratic constant 2nd difference

6.

x	y
0	70
1	75
2	80
3	85
4	90

Linear constant 1st difference

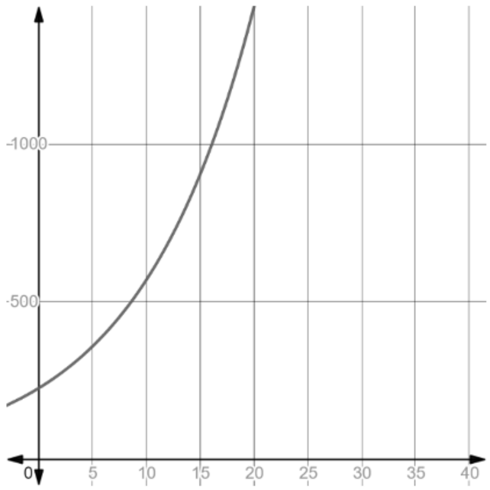
7.

x	y
0	225
1	247
2	271
3	297
4	326

Exponential, no constant 1st or 2nd difference

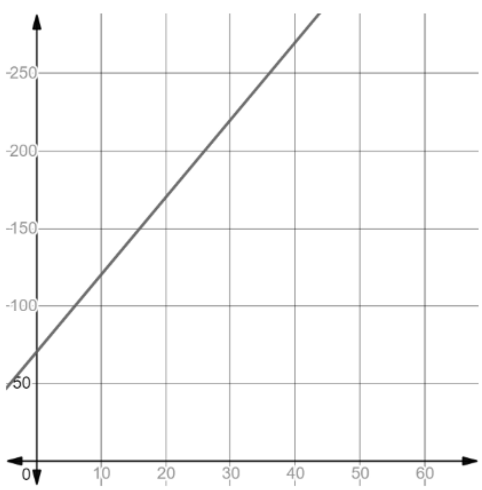
Given a graph, identify the function as linear, quadratic, or exponential. Explain your choice.

8.



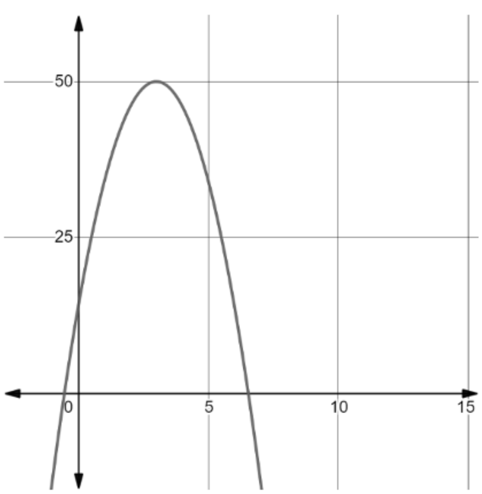
Exponential
-constant ratio from one y-value to the next.
-follows exponential curve.

9.



Linear
-straight line
-constant slope

10.



Quadratic
-follows quadratic curve
-constant second difference in y-values.

Algebra 1 Name_Answer Key_________
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The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502

1. Determine the average rate of change from 2010 to 2013.

$(2010, 76743)$ $\frac{y_2-y_1}{x_2-x_1}$ $\frac{103{,}502-76{,}743}{2013-2010}$ $\frac{26{,}759}{3}$ $\approx \boxed{8{,}919.67}$

$(2013, 103502)$

2. Interpret the average rate of change for the interval 2010 to 2013 in terms of the context.

Between the years 2010 to 2013 the amount of blueberries imported increased $8{,}919.67$ tons per year, on average.

3. Determine the average rate of change from 2005 to 2009.

$(2005, 26319)$ $\frac{y_2-y_1}{x_2-x_1}$ $\frac{60{,}450-26{,}319}{2009-2005}$ $\frac{34{,}131}{4}$ $\boxed{= 8{,}532.75}$

$(2009, 60450)$

4. Interpret the average rate of change for the interval from 2005 to 2009 in terms of the context.

Between the years of 2005 and 2009, an average of $8{,}532.75$ more tons of blueberries were imported each year.

5. How does the average rate of change for the interval from 2005 to 2009 compare to the average rate of change for the interval 2010 to 2013?

The average rate of change from 2005-2009 is fewer by 386.92 compared to 2010-2013.

The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.

6. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1980 to 1985?

$1980 - 1970 = 10$

$1985 - 1970 = 15$

$5(10)^2 - 160(10) + 6{,}179$

$500 - 1{,}600 + 6{,}179$

$-1{,}100 + 6{,}179$

$= 5{,}079$

$(10, 5079)$

$5(15)^2 - 160(15) + 6{,}179$

$1{,}125 - 2{,}400 + 6{,}179$

$-1{,}275 + 6{,}179$

$= 4{,}904$

$(15{,}4904)$

$\frac{y_2 - y_1}{x_2 - x_1}$

$\frac{4{,}904 - 5{,}079}{15 - 10}$

$\frac{-175}{5}$ $\boxed{= -35}$

7. Interpret the average rate of change for the interval 1980 to 1985 in terms of the context.

From 1980 to 1985 the average rate of change in regards to passengers traveling by train was -35 million miles. So, on average there was a decrease in train travel during this time period.

8. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1985 to 1990?

$1985: (15{,}4904)$

$1990: 1990 - 1970 = 20$

$5(20^2) - 160(20) + 6{,}179$

$2{,}000 - 3{,}200 + 6{,}179$

$-1{,}200 + 6{,}179$

$= 4{,}979$

$(20, 4979)$

$\frac{y_2 - y_2}{x_2 - x_1}$

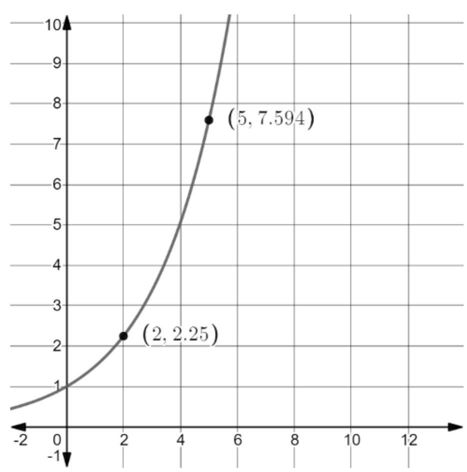
$\frac{4{,}979 - 4{,}904}{20 - 15}$

$\frac{75}{5}$ $\boxed{= 15}$

9. Interpret the average rate of change for the interval 1985 to 1990 in terms of the context.

From 1985 to 1990 the average rate of change in regards to passengers traveling by train was 15 million miles. So, on average there was an increase in train travel during this time period.

10. The graph of a function is shown. Points $(2{,}2.25)$ and $(5{,}7.594)$ are on the function. Calculate the average rate of change for the line that passes through the points $(2{,}2.25)$ and $(5{,}7.594)$.



$\frac{y_2 - y_1}{x_2 - x_1}$ $\frac{7.594 - 2.25}{5 - 2}$ $\frac{5.344}{3}$ $\approx 1.781\overline{3}$

Algebra 1
Unit 13
Lesson H1 Practice

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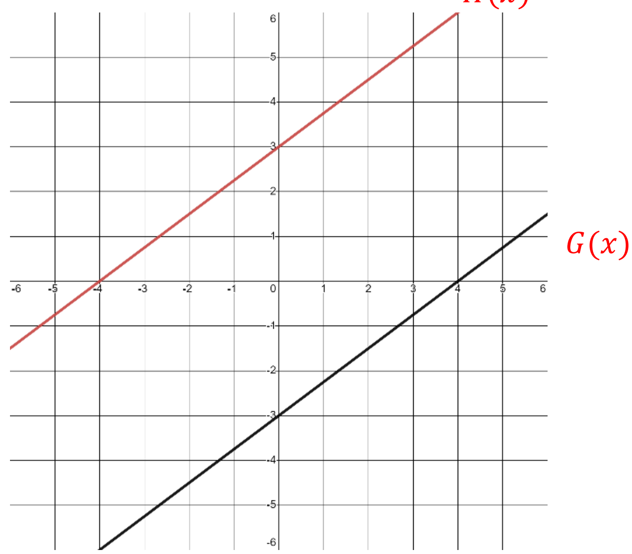
The graph of the function $G(x)$ is shown.



1. Write the function, $G(x)$, that represents the line.

   $G(x) = \frac{3}{4}x - 3$

2. Draw a line that is parallel to $G(x)$ with a y-intercept of 3. Label this line $H(x)$.

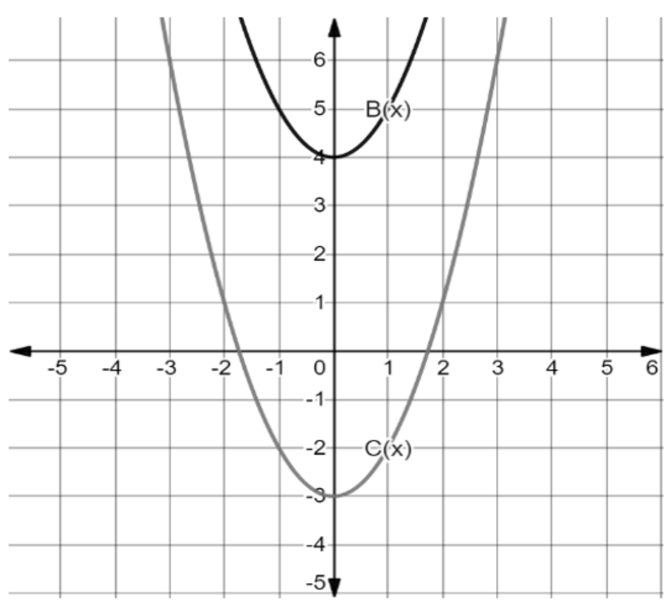
3. Describe how $G(x)$ can be shifted to coincide with $H(x)$.

   Shift up 6 units

4. Write the function $H(x)$ in terms of $G(x)$.

   $H(x) = G(x) + 6$

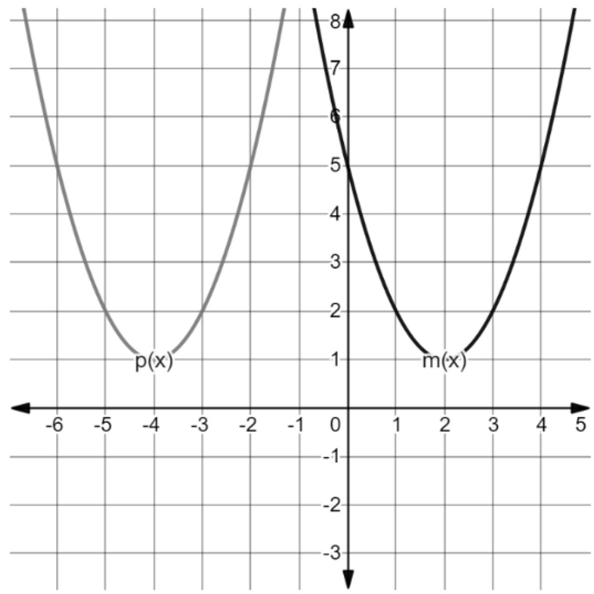
5. The graphs of $B(x)$ and $C(x)$ are shown on the graph.



Write the transformation by completing the equation.

$C(x) = B(x) + (-7)$

The functions $m(x)$ and $p(x)$ are shown on the graph.



6. Write $m(x)$ in vertex form.

$m(x) = (x - 2)^2 + 1$

7. Write $p(x)$ in vertex form.

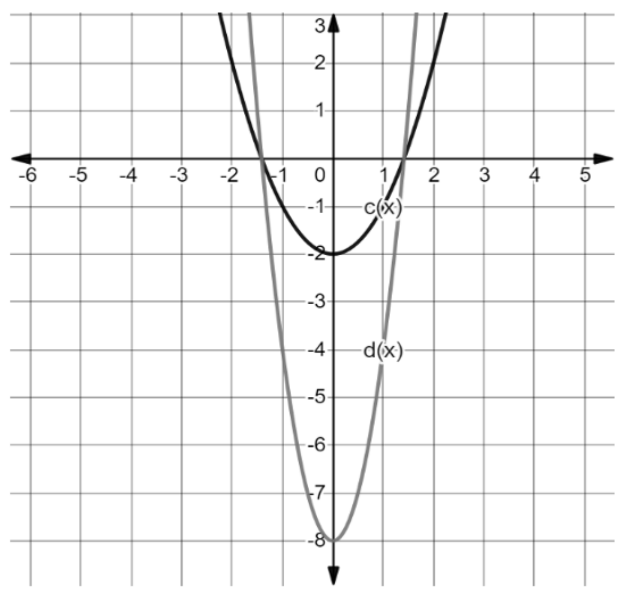
$p(x) = (x + 4)^2 + 1$

8. Describe how $m(x)$ was transformed to create $p(x)$.

Shifted left 6 units

9. The functions $c(x)$ and $d(x)$ are shown on the graph, where $d(x)$ is a transformation of $c(x)$. What is the value of k such that $d(x) = kc(x)$?

$k = 4$



10. The table of values for $d(x)$ and $f(x)$ are shown, where $f(x)$ is a transformation of $d(x)$. Write a description of the transformation.

$y = d(x)$		$y = f(x)$	
x	y	x	y
-2	16	-2	20
-1	4	-1	8
0	0	0	4
1	4	1	8
2	16	2	20

$d(x)$ is shifted up 4 units to form $f(x)$.

Algebra 1
Unit 13
Lesson H2 Practice

Name_Answer Key________
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Use the functions $b(x) = -2x^2 + 6x - 3$ and $c(x) = 8 + 3x$ to determine each new function. Write your answer in standard form.

1. $(b + c)(x)$

$$\begin{array}{r} = -2x^2 + 6x - 3 \\ \underline{+ \qquad\quad 3x + 8} \end{array}$$

$$\boxed{(b + c)(x) = \quad -2x^2 + 9x + 5}$$

2. $(b - c)(x)$

$$= (-2x^2 + 6x - 3) - (8 + 3x)$$

$$= -2x^2 + 6x - 3x - 3 - 8$$

$$\boxed{(b - c)(x) = \quad -2x^2 + 3x - 11}$$

3. $(c - b)(x)$ $= (8 + 3x) - (-2x^2 + 6x - 3)$

$$= 8 + 3x + 2x^2 - 6x + 3$$

$$\boxed{(c - b)(x) = \quad 2x^2 - 3x + 11}$$

4. $(b \cdot c)(x)$

$$= (-2x^2 + 6x - 3)(8 + 3x)$$

$$= -16x^2 + 48x - 24 + (-6x^3) + 18x^2 - 9x$$

$$\boxed{(b \cdot c)(x) = -6x^3 + 2x^2 + 39x - 24}$$

5. $\left(\frac{b}{c}\right)(x)$

$$\boxed{= \frac{-2x^2 + 6x - 3}{3x + 8}}$$

6. Write the restriction on the domain for $\left(\frac{b}{c}\right)(x)$.

$$3x + 8 \neq 0$$

$$3x \neq -8$$

$$x \neq -\frac{8}{3}$$

$$\boxed{x \neq -\frac{8}{3}}$$

Paulina and Stefan are renting a boat for a trip to the Keys. The boat costs $\$200$ per day. The rental includes 125 miles for free and then charges $\$0.50$ per mile. The function $B(m) = 0.50(m - 125) + 200$ models the cost of the boat, B, in terms of miles traveled, m. They determine gas will cost $\$1.75$ for each mile they travel. The company charges a $\$75$ fee for the first tank of gas. The function $F(m) = 1.75m + 75$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $B(m)$?
   Answers may vary.

   Must include $x \geq 0$, may include a reasonable maximum as well.

8. What is a reasonable domain for $F(m)$?
   Answers may vary.

   Must include $x \geq 0$

9. Determine the function that gives the total cost, $C(m)$, in dollars, based on the miles traveled.

$$C(m) = B(m) + F(m)$$

$$= 0.50(m - 125) + 200 + 1.75m + 75$$

$$= 0.50m - 62.5 + 1.75m + 275$$

$$\boxed{C(m) = 2.25m + 212.50}$$

10. What is a reasonable domain for $C(m)$?
    Answers may vary.

    Must include $x \geq 0$